HEALTH ANALYSIS

Healthcare data, turned into decisions.

Two years of patient visit and cost data

110,300 visits · $276M total cost · 7 diagnoses · 5 doctors

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01 · The brief

What is the cost of care telling us that operations alone cannot?

The question driving this work:

Where is cost concentrated, which diagnoses drive volume, how is workload distributed across doctors, and what patterns in age and gender shape the patient mix? Operations sees the schedule. Finance sees the invoices. Neither sees the full picture. Bringing the data into one model surfaces decisions that neither view can make alone.

What sits underneath the dashboard

SOURCE

Raw Data

114,000 rows of patient visit records

9 columns: visit date, patient ID, age, gender, diagnosis, treatment, doctor, cost, age range

Mixed-case categorical values, inconsistent date formats, missing doctor and diagnosis fields

CLEANED

Clean Data

110,300 rows after deduplication and validation

11 columns including derived Age Group and Month

3,700 rows removed for invalid dates or null IDs (3.2 percent of source)

Unassigned doctor and Unknown diagnosis flags kept and quarantined for review

The picture in six numbers

$276M

Total cost across 2 years

110,300

Patient visits processed

$2,501

Mean ticket per visit

5,000

Unique patients seen

7

Diagnosis categories tracked

22 visits

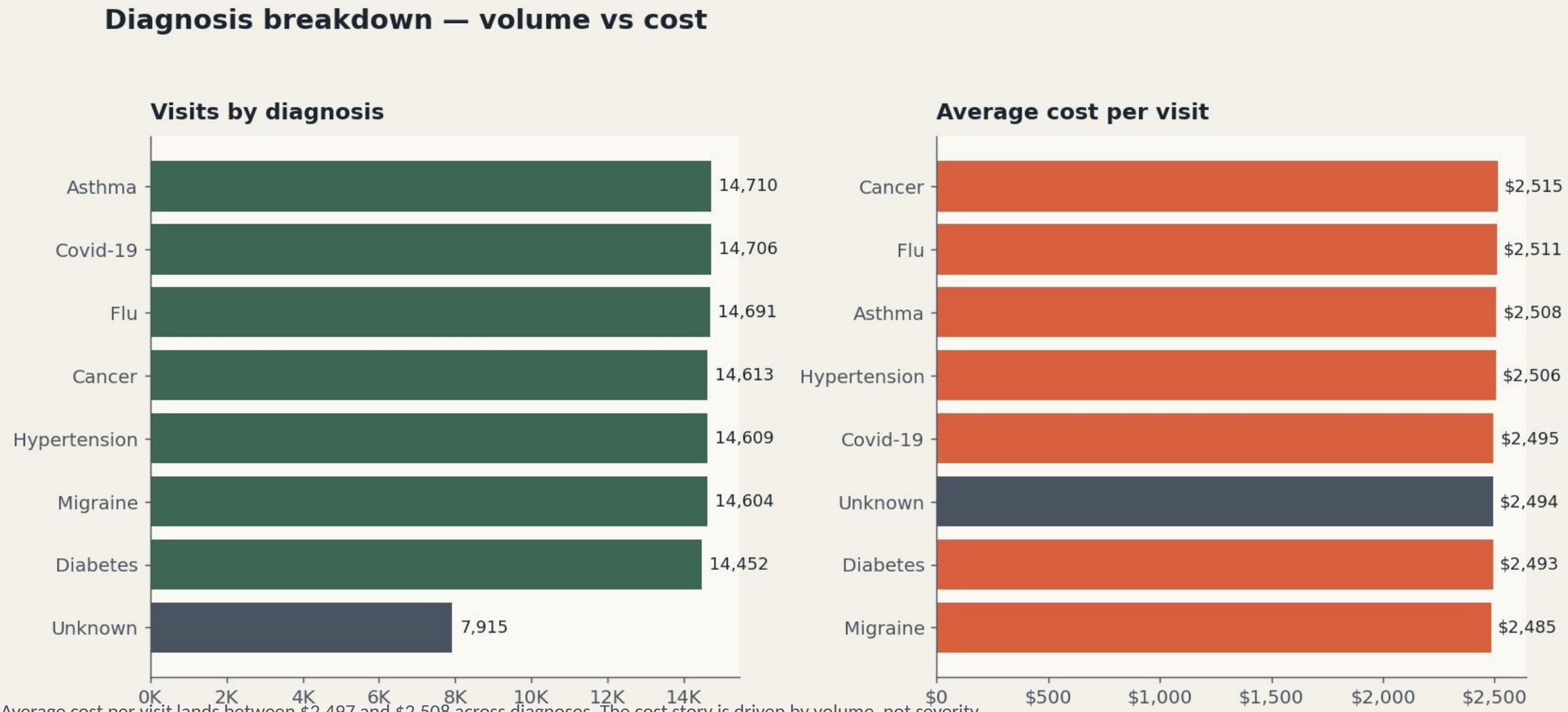
Average visits per patient

Cost is evenly spread, with one exception



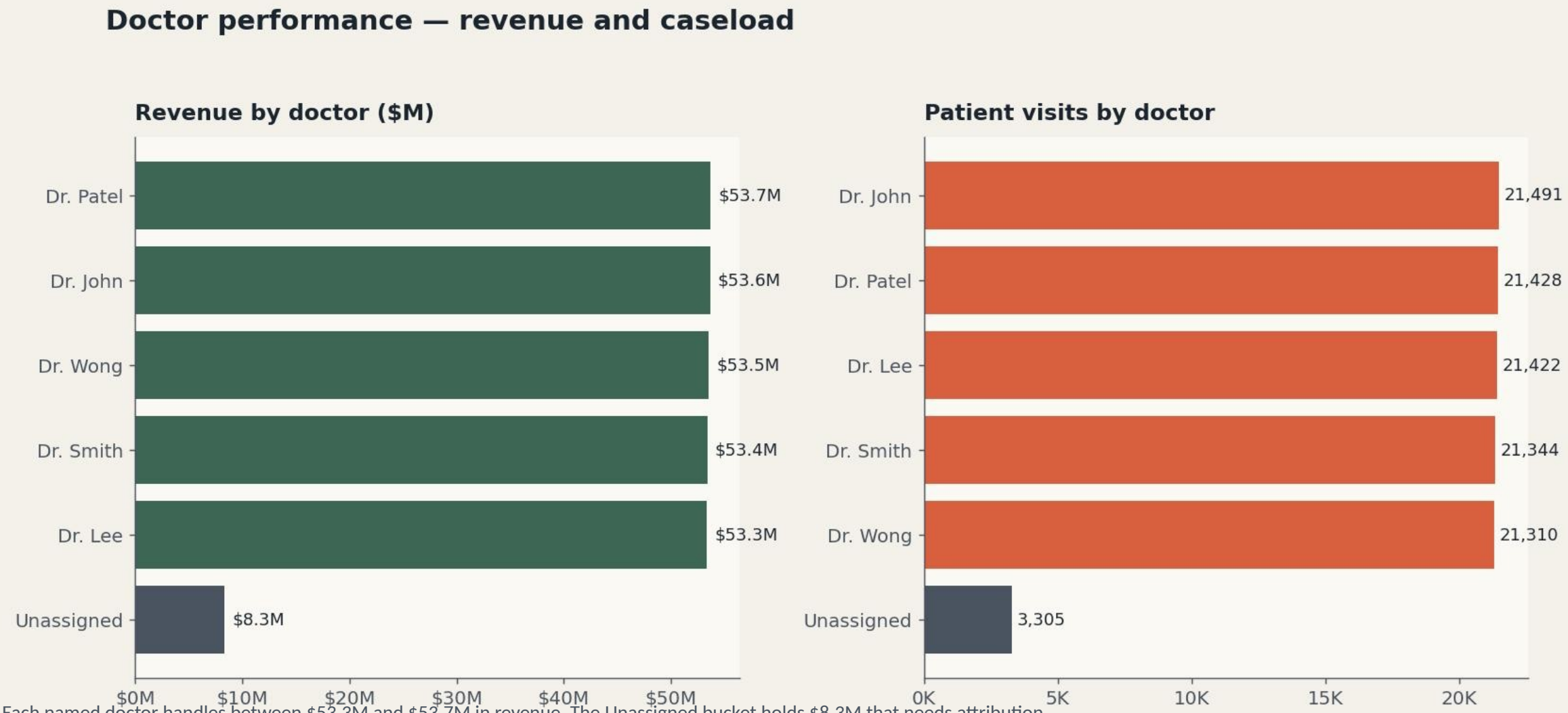
Six of seven named diagnoses sit within a $1M band. The Unknown category alone shows a $17M gap to the leaders.

Volume and ticket size move together



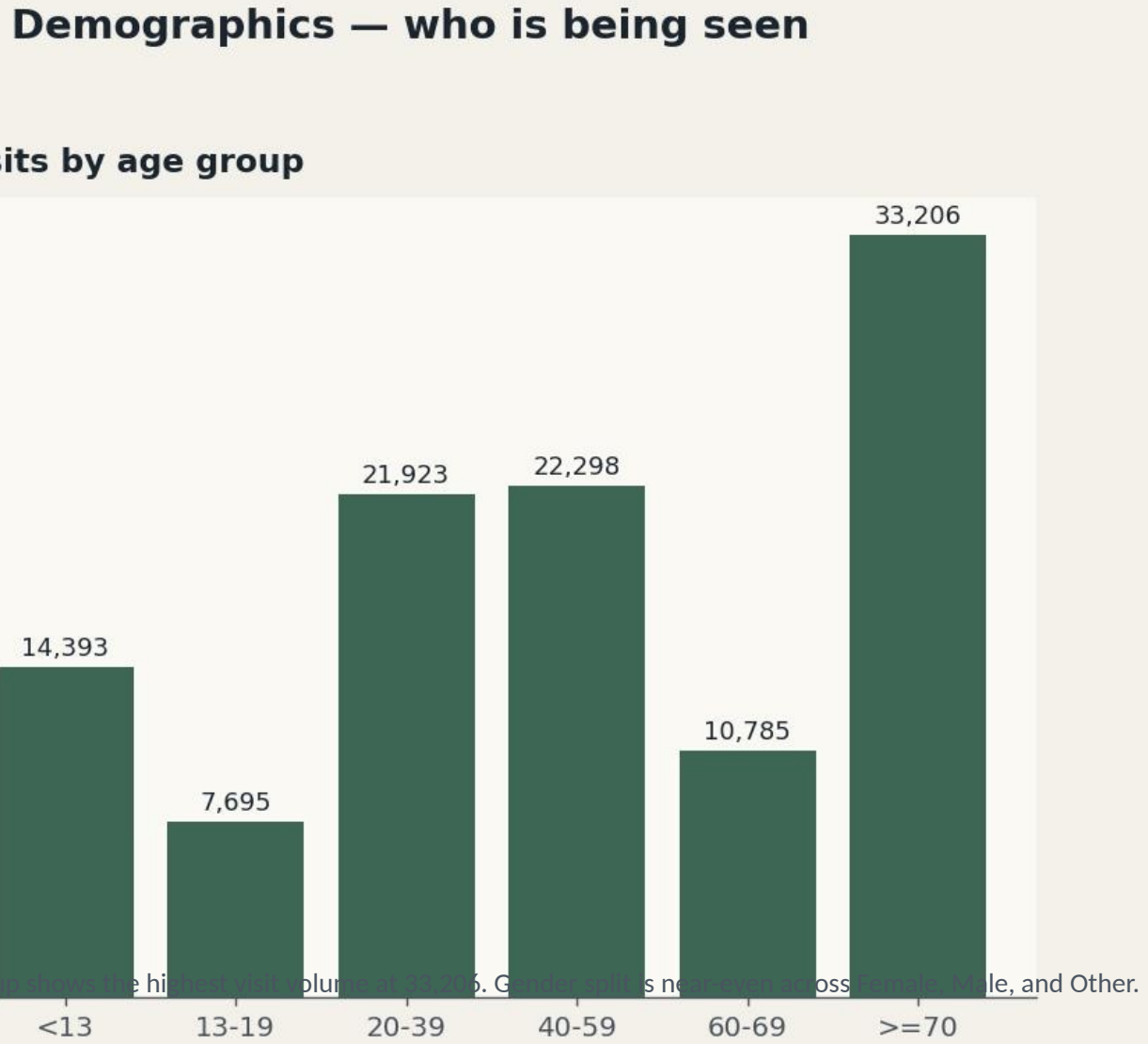
Average cost per visit lands between $2,497 and $2,508 across diagnoses. The cost story is driven by volume, not severity.

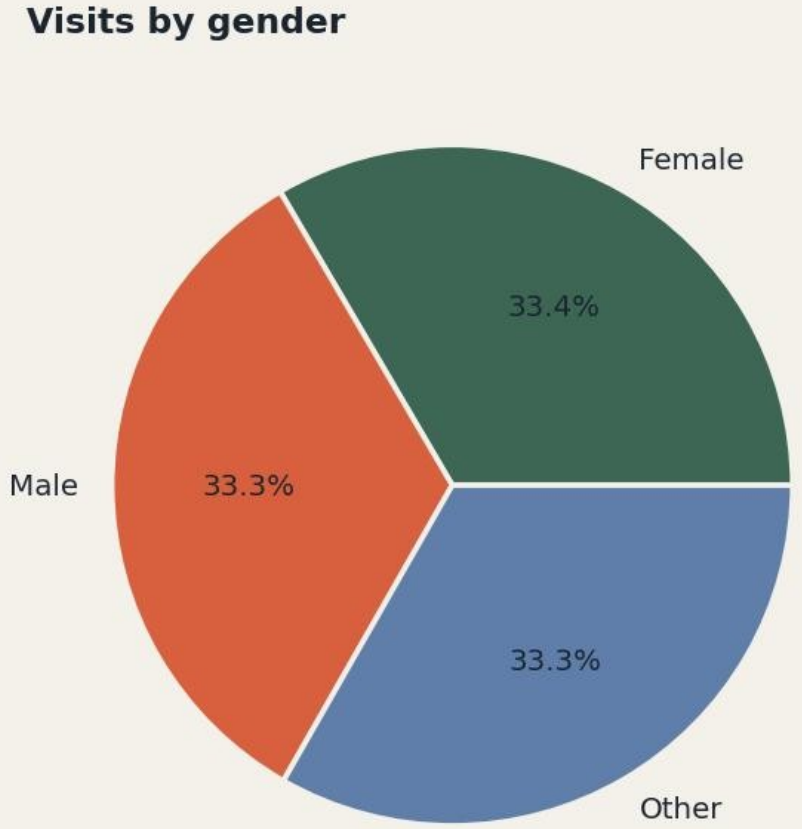
Caseload is balanced across the five named doctors



Each named doctor handles between $53.3M and $53.7M in revenue. The Unassigned bucket holds $8.3M that needs attribution.

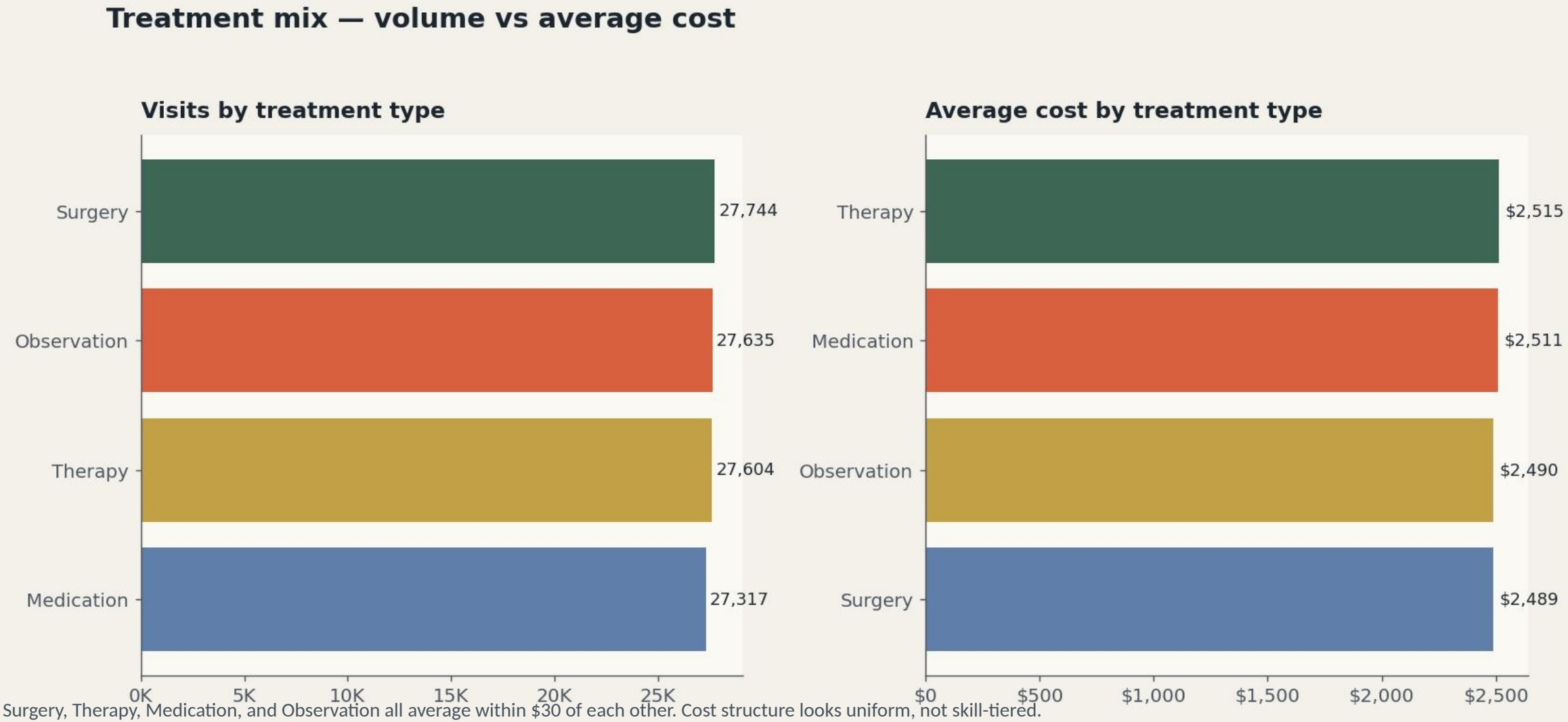
Who is being seen





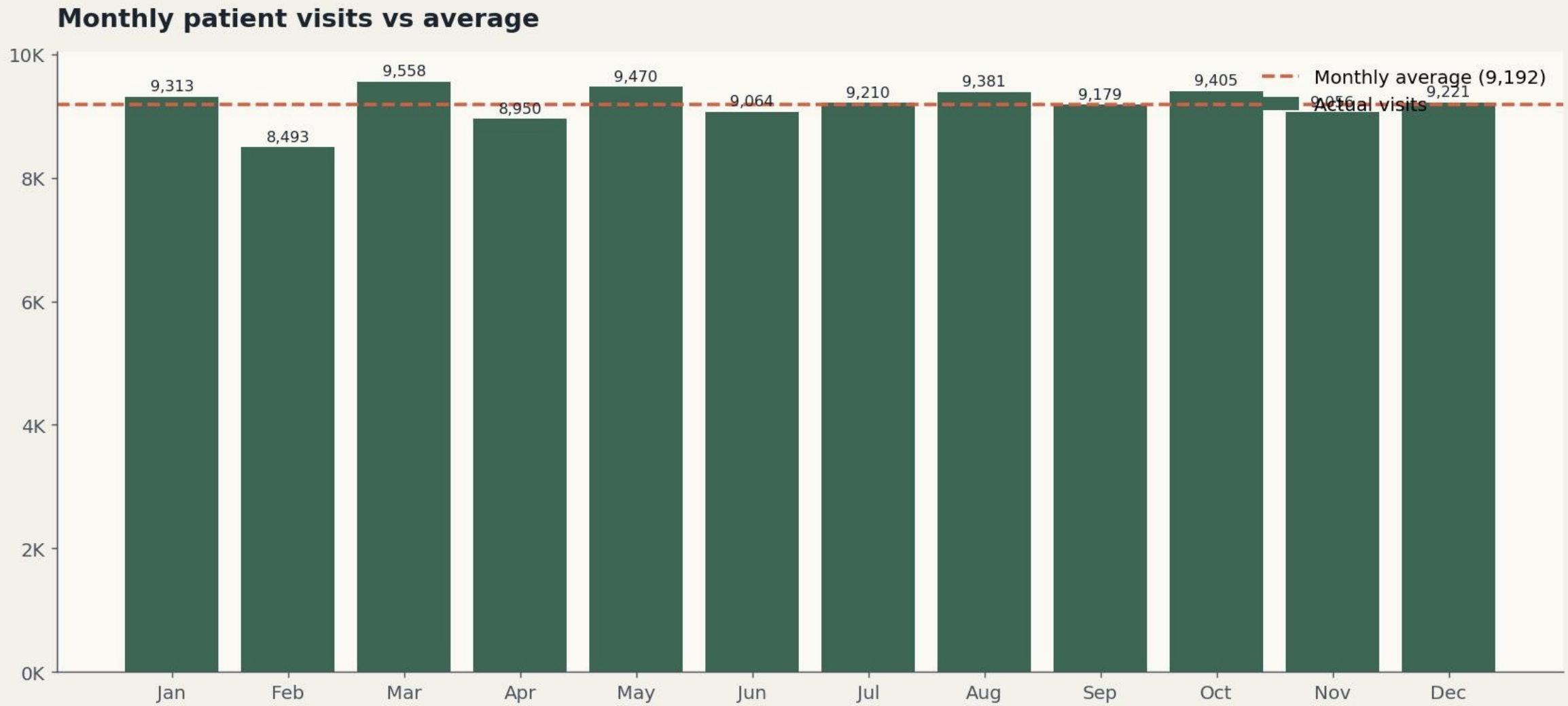
The 70+ group shows the highest visit volume at 33,206. Gender splits near-even across Female, Male, and Other.

Four treatment types, four near-identical averages



Surgery, Therapy, Medication, and Observation all average within $30 of each other. Cost structure looks uniform, not skill-tiered.

No clear seasonality in visits or revenue



Monthly visits hover within 10 percent of the 9,192 monthly average. Capacity planning can run on a flat baseline.

What the numbers say

01 Cost spreads almost evenly across named diagnoses

Six of seven diagnoses cluster in a $36M to $37M band. Resource planning does not need to favour one disease group.

02 Unknown diagnosis is the largest data quality gap

$19.7M of cost sits against unlabelled diagnoses. Fixing intake categorization unlocks visibility into roughly 7 percent of the book.

03 Doctor workload is balanced, with one anomaly

Five named doctors carry near-equal revenue. The Unassigned bucket holds $8.3M that needs attribution to a real provider.

04 The 70+ group is the largest patient cohort

33,206 visits, the highest of any age bracket. Service capacity planning should anchor on this group.

05 Treatment type does not predict cost

Surgery and observation average within $30 of each other per visit. Cost variance lives elsewhere, likely in volume and case complexity.

What to do next

01 Fix the intake categorization process

Unknown diagnoses are $19.7M of invisible spend. A required-field rule at intake closes the gap.

High

02 Resolve the Unassigned doctor records

Audit the 3,289 visits with no doctor. Reassign or archive to remove the $8.3M attribution gap.

High

03 Move from monthly to weekly cost monitoring

Monthly variance is too coarse for early warnings. Weekly tracking surfaces shifts inside the cycle.

Medium

04 Build a doctor-level utilization view

Balanced revenue does not mean balanced effort. Add hours, visit counts per shift, and patient outcomes.

Medium

05 Validate the 70+ cohort capacity plan

This group drives the largest visit volume. Confirm the schedule, beds, and specialist mix match the load.

Low

How it was built

Excel	Data cleaning, pivot tables, formulas, and dashboard layout
Power Query	Date normalization and categorical standardization
Pivot Tables and Charts	Aggregation by diagnosis, doctor, age group and month
Slicers	Cross-filter dashboard by gender, diagnosis, and time period
Conditional Formatting	Visual emphasis on outliers and target comparisons

CLOSING

When the foundation is right, the dashboard is the easy part.

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